

Final Project Stat 419

100 points

Instructions: Answer the questions below using SAS. Do include relevant charts and tables from SAS and discuss them. Do not include any SAS output that is not directly relevant to your discussion. When reasonable, compile SAS results into easily readable tables. Your final product should be in the format of a formal paper (complete sentences, edited for grammar, etc.).

The data set concerns survival time of patients with advanced, inoperable lung cancer. Variables include whether or not the patient received chemotherapy, the type of cancer, how long the patient survived (in days), whether or not the patient's survival time is censored, Karnofsky score (see additional documentation), months from diagnosis, age in years, and whether or not the patient received any prior therapy.

1. Exploratory Data Analysis (12 points):

Provide descriptive statistics for the quantitative variables, and summary information for the categorical variables (counts/percentages). For each variable, provide a brief (1-2 sentence) description (you can include means, medians, min, max, etc.). Please create a summary table or tables to include in your report (*do not copy/paste SAS output*).

2. Survival Analysis (40 points):

Create an overall survival curve for these patients and include it in your report. Report the median survival time (use SAS output for this, do not estimate from the graph).

Are the survival functions significantly different between:

- Patients who received chemotherapy and those who did not?
- Patients who have different types of cancer?
- Whether or not patients received prior treatment?
- Number of months from diagnosis (stratify by using 1 year as a cutoff)
- Age groups? (stratify by using age 60 as a cutoff)
- Karnofsky scores? (stratify by using 50 as a cutoff, then again by using Karnofsky groups: 0-50, 50-80, 80-100)

Report median survival times for all groups in every case, and include graphs of the survival curves. If there are statistically significant differences between groups, report p-values and comment on which group or groups had better survival times. If there are not statistically significant differences between groups, mention that.

3. Advanced Survival Analysis (20 points):

Check to see if there are differences in survival curves between the treatment groups, while simultaneously looking to see if there are differences between:

- Age
- Karnovsky score
- Cancer type
- Number of months from diagnosis

Use the Tukey adjustment method for multiple comparisons. Use the same stratifications as you did in problem 2. Comment on significant differences (be specific about which strata are different from others, if any, and include p-values if appropriate). Include relevant survival curves.

4. Proportional Hazard Regression (20 points):

Use SAS to model survival time as a function of the other covariates. Start with all covariates in the model (call it Model 1).

Remove all variables except the three most significant (call it Model 2). Discuss the statistically significant covariates, and interpret the estimated hazard ratios for all statistics.

Create a table that contains variable coefficient estimates with p-values and a model-fit criterion (see sample table below for guidance).

Model	-2LogL	Var 1 (p-value)	Var 2 (p-value)	Var 3 (p-value)	Var 4 (p-value)	Var 5 (p-value)	Var 6 (p-value)
Model 1	132	1.34 (<0.001)	1.34 (<0.001)	1.34 (<0.001)	1.34 (<0.001)	1.34 (<0.001)	1.34 (<0.001)
Model 2	102	2.54 (<0.001)		2.54 (<0.001)	2.54 (<0.001)		

5. Summary (8 points):

In a paragraph, summarize your findings. What variables have the strongest impact on survival? Which variables had little to no impact? What are your conclusions from the regressions?

6. Bonus (10 points)

Two of the types of cancer are much more deadly than the other two. Which two types are most deadly? Explore whether treatment is more effective in the two less-deadly cancer types versus the two more-deadly cancer types. Justify your approach and support it with SAS output.